

TribalScholar AI

Brainstorming Mind Map and Idea Bank

Central Idea: A configurable, AI-assisted scholarship and fellowship management platform for Scheduled Tribe education schemes. It enables authorised officials to change scheme forms, documents, eligibility rules, workflow and scoring without code changes. AI assists verification; humans make final decisions.

Central mind map

Branch	Brainstorming ideas
1. Users	Applicant; institute verifier; district/state nodal officer; scheme officer; selection committee; finance/DBT officer; super admin; auditor; CSC/facilitation-centre operator.
2. Schemes	Pre-Matric; Post-Matric; Top Class Education; National Fellowship; National Overseas Scholarship; future state-specific schemes.
3. Applicant journey	Register → OTP → profile → preliminary scheme recommendation → dynamic form → documents → OCR confirmation → submit → status/deficiency → review → decision → appeal/grievance.
4. Configurator	Add fields; documents; rules; deadlines; seats; score weights; workflow; category limits; publish versions; simulate changes.
5. AI support	OCR; document type; completeness; blur/crop/readability; field extraction; name matching; income/DOB consistency; duplicate possibility; review priority.
6. Governance	Human review; audit trail; evidence-based decisions; applicant explanation; privacy; fairness monitoring; appeal process.
7. Inclusion	Hindi/English; simple language; mobile-first; PWA; offline draft; compressed upload; retry; large buttons; screen-reader labels; voice input roadmap.
8. Technology	React/Vite; FastAPI; PostgreSQL; MinIO; OpenCV/OCR; Redis queue; Docker; charts; tests.

Feature idea bank

Area	Must-have ideas	Advanced ideas
Applicant portal	Profile, dynamic form, document upload, status timeline, deficiencies, notifications.	Voice guidance, chatbot, offline-first PWA, facilitation-centre assisted flow.
Document verification	File validation, preview, checklist, OCR extraction confirmation.	Quality score, wrong-document detection, expiry check, document similarity.
Eligibility	Configured deterministic rules and reason messages.	Rule version comparison, what-if eligibility preview, policy simulation.
Selection	Configurable merit components and score breakdown.	Seat simulation, waitlist logic, committee conflict-of-interest declaration.
Admin	Scheme CRUD, field builder, document builder, rule builder, dashboard.	Scheme templates, approval workflow for configuration publishing, impact analysis.
Trust	Audit timeline, role scope, reasons for decisions.	Fairness dashboard, tamper-evident audit hash chain, decision-evidence packet.

Uniqueness brainstorming

1. No-code policy engine: Officials configure the scheme rather than developers hard-coding each scheme.
2. Human-in-the-loop AI: AI flags possible issues and explains why; officers decide. No automatic fraud labels, rejection or selection.
3. Tribal-document-aware matching: Spelling/transliteration variation goes to manual review rather than automatic failure.

4. Evidence-first decisions: Each decision has rule result, document evidence, officer rationale and timeline entry.
5. Low-bandwidth inclusion: Offline draft, small uploads, resume after failure and SMS-style updates for remote areas.
6. Fairness monitoring: Aggregate checks for disproportionate deficiency/rejection rates or OCR errors across permitted groups.
7. Scheme simulator: Admin changes draft thresholds, score weights or seats and previews the potential impact before publishing.

Problem → solution ideas

Pain point	Idea	Demo moment
Every scheme is different	Versioned scheme configuration	Add a document/rule from dashboard; application form updates.
Manual document screening	OCR plus completeness queue	System identifies missing admission letter before officer review.
Unclear applicant communication	Specific deficiency templates	Applicant sees exact missing document and deadline.
Possible unfair rejection	Name variation advisory flag	“Hembram” vs “Hembrom” receives manual review, then officer approval.
Limited transparency	Status timeline and audit log	Show actor, time, action, reason and evidence for each transition.
Policy changes affect outcomes	Scheme simulator	Change academic weight and show simulated ranking movement.

AI idea canvas

AI capability	Inputs	Output to user/officer	Human safeguard
OCR extraction	PDF/image document	Structured fields and confidence	Applicant confirms; officer verifies.
Quality check	Image clarity/crop/text density	Re-upload recommendation	Never reject solely for quality flag.
Consistency check	Profile + document fields	Match/variation/mismatch evidence	Borderline and mismatch cases reviewed.
Duplicate possibility	Approved signals and file hashes	Contributing indicators and review priority	No “fraud confirmed” label or auto-rejection.
Document classifier	Document text/layout	Likely document type	Officer can override classification.
Fairness monitor	Aggregated outcomes and OCR errors	Trend indicators	Use for review, not individual adverse decisions.

Brainstormed screens

Landing • Registration/OTP • Profile • Scheme recommendation • Scheme details • Dynamic form • Document upload/preview • OCR confirmation • Applicant dashboard • Deficiency correction • Appeal/grievance • Officer queue • Verification workspace • Committee review • Finance/sanction • Admin dashboard • Scheme configurator • Rule builder • Audit log • Fairness dashboard • Simulator.

Demo story idea

Use synthetic applicant Laxmi Hembram from Jharkhand, ST category, PhD, income INR 240,000. The system recommends a configured fellowship as preliminary guidance. OCR extracts certificate and income details. Her marksheets says “Laxmi Hembrom”; AI reports a possible spelling variation with evidence. It does not reject the application. The officer reviews and approves the document, the merit score is shown, the applicant timeline updates and the audit record captures every action.

MVP priority brainstorm

Priority	Features
P0 — demo essentials	Authentication/RBAC, applicant profile, configurator, dynamic form, document upload, deterministic eligibility rules, workflow, officer action, audit timeline.
P1 — AI proof	OCR mock/real extraction, completeness checker, name/income/DOB comparison, manual-review flags, explainable risk indicators.
P2 — judge differentiators	Fairness dashboard, scheme simulator, offline draft, bilingual UI, appeal/grievance, analytics.
Future	Official integrations after approvals, regional languages, advanced OCR models, production security operations, DBT adapters.

Guardrails

Never claim 100% OCR accuracy, automatic fraud detection, automatic candidate selection, real Aadhaar/DBT integration, immediate government production use, or replacement of officials. Say: AI-assisted verification, configurable prototype, risk-based prioritisation, human approval and future official integration subject to approval.

Prompt for an AI builder

Read this brainstorming file together with the full project blueprint. Convert the P0 features into a working vertical-slice MVP using React/Vite/TypeScript, FastAPI/Python, PostgreSQL and Docker. Keep rules configuration-driven and versioned. Implement AI only as advisory evidence. Add tests for RBAC, applicant isolation, workflow transitions, upload security and audit logs. Use synthetic data only.